Task 5 — Dynamic Quoting Under Inventory Pressure

1. Approach and Functional Form

The quoting function produces bid and ask half-spreads (δ_b, δ_a) decomposed into an *inventory skew* and a *symmetric spread*:

$$\delta_b = \underbrace{+\kappa\,\sigma\,I\,(1+\beta_{\text{eod}}\,\eta^2)}_{\text{inventory skew}} \;+\; \underbrace{W\cdot\Big(\frac{\gamma_s\,\sigma}{2}+\beta_\alpha\,\sigma\,\alpha+\beta_\eta\,\sigma\,\eta^2\Big)}_{\text{symmetric spread}},$$

$$\delta_a = \underbrace{-\kappa\,\sigma\,I\,(1+\beta_{\text{eod}}\,\eta^2)}_{\text{inventory skew}} \;+\; \underbrace{W\cdot\Big(\frac{\gamma_s\,\sigma}{2}+\beta_\alpha\,\sigma\,\alpha+\beta_\eta\,\sigma\,\eta^2\Big)}_{\text{symmetric spread}},$$

clipped to the constraint $[c_{\min}\cdot\sigma,\ 50\,\text{bps}]$ with $c_{\min}=0.5$.

The **inventory skew** shifts quotes against the current position: a long inventory pushes the bid up and the ask down, encouraging inventory reduction before the end-of-day penalty (Eq. 16). The term $\beta_{\text{eod}}\,\eta^2$ amplifies this urgency as session-end approaches ($\eta\to 1$). The **symmetric spread** $W\cdot(\cdots)$ compensates for flow risk: it widens with volatility σ, the Task-3 adversity score α, and session progress η.

The **regime multiplier** $W \geq 1$ adapts to abrupt market shifts via a two-timescale EWMA on (σ,α) with half-lives of 100 and 5 000 trades:

$$W \;=\; \min\!\Big(2.5,\;\; 1+\max\!\big(0,\,\tfrac{\hat\sigma_{\text{fast}}}{\hat\sigma_{\text{slow}}}-1\big)\cdot 0.6+\max\!\big(0,\,\tfrac{\hat\alpha_{\text{fast}}}{\hat\alpha_{\text{slow}}}-1\big)\cdot 0.4\Big).$$

W is one-sided: it only widens above the static optimum, never tightens below it.

2. Parameterization Intuition & Validation Results

Parameters are chosen by minimising the negative log-mean Sharpe across a 4×4×4 grid of hidden (λ,γ,ϕ) values via Differential Evolution, using an analytic evaluator (≈5 ms/call) in place of Monte Carlo.

Param	Value	Intuition
κ	0.00114	Skew intensity; kept small to avoid killing fill rates
γ_s	1.648	Base spread as $\times\,\sigma$; volatility-proportional floor
β_α	1.642	Adversity loading; widens on informed-flow signal
β_η	0.282	Session-progress widening; manages end-of-day risk
β_{eod}	0.070	EOD skew amplifier; urgently flattens inventory

Table 1: Optimised parameters and their roles.

Split	Med. PnL	Med. Sharpe	Worst Sharpe
Train (n=25 days)	$24,522	147.8	18.7
Val. (n=8 days)	$ 7,586	77.1	9.6
Test (n=9 days)	$ 8,991	67.7	3.2

Table 2: 8 corner scenarios per split. All 24 scenario-split pairs yield positive PnL. Worst case: λ=0.3, γ=4, ϕ=0.5 (test, $23, Sharpe 3.2). Moderate train-to-test decay confirms robustness.

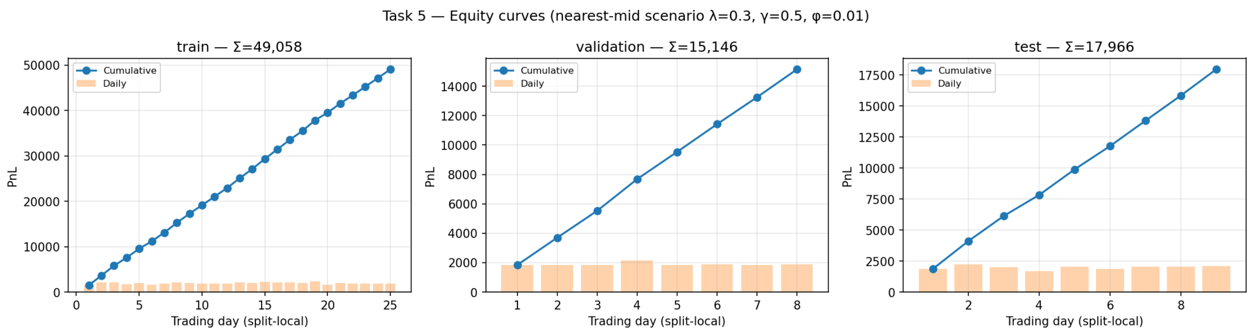

Figure 1: Daily PnL (bars) and cumulative PnL (line) for the representative scenario (λ=0.3, γ=0.5, ϕ=0.01) across all three splits. Monotone cumulative growth with no sustained drawdown confirms stable quoting behaviour across changing market conditions.